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Ethical Aspects of
Artificial Intelligence:

Bias in Machine Learning

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Overview

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What is Bias in Machine Learning?

- Bias refers to systematic unfairness in ML systems
- Occurs when models produce unequal outcomes for different groups
- Often caused by biased or incomplete training data
- ML systems may amplify existing social inequalities

Examples of Bias in ML Systems

- COMPAS algorithm used in US criminal sentencing
- Higher false-positive rates for Black defendants
- Facial recognition systems show lower accuracy for darker skin tones
- Bias observed in hiring and credit scoring algorithms

Why does Bias Occur in ML?

01 Historical discrimination reflected in training data:
Past social inequalities are often embedded in data, which ML models may learn and repeat.

02 Imbalanced or non-representative datasets:
Some groups appear less frequently in data, leading to worse model performance for them.

03 Human bias introduced during data collection:
Bias can enter through subjective decisions in data labeling, selection, or preprocessing.

04 Lack of transparency in ML models “black boxes”:
Complex models are difficult to interpret, making biased behavior hard to detect.

Societal Impact of ML Bias

Hypotheses

If biased data and opaque models are used in AI systems, automated decisions will likely reinforce social inequalities and reduce public trust.

- **Discriminatory automated decisions**
  Biased AI systems can lead to unfair decisions in areas such as hiring, credit scoring, or criminal justice, disproportionately affecting certain groups.
- **Reduced trust in AI systems:**
  When people experience or perceive unfair AI decisions, confidence in automated systems and their adoption decreases.
- **Reinforcement of social inequalities:**
  Biased AI can reproduce and strengthen existing social and economic inequalities over time.
- **Ethical and legal concerns:**
  The deployment of biased AI raises ethical questions and may conflict with fairness, accountability, and anti-discrimination laws.

How Can Bias Be Reduced?

01 **Use diverse and representative datasets:**
Ensure data reflects all demographics and scenarios to prevent biased outcomes.

02 **Perform regular bias audits and testing:**
Continuously monitor and evaluate models for unfair patterns using tools like IBM’s AI Fairness 360.

03 **Increase transparency and explainability:**
Make model decisions understandable to detect bias and build trust.

04 **Maintain human oversight in critical decisions:**
Humans should review high-stakes decisions to ensure accountability and prevent harm.

05 **Apply ethical and legal frameworks:**
Follow ethical principles and regulations, such as the EU AI Act, to guide responsible AI use.

Sources

Slide 3

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- IBM – AI Fairness 360 Toolkit
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Thank You

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Introduction to Artificial Intelligence
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